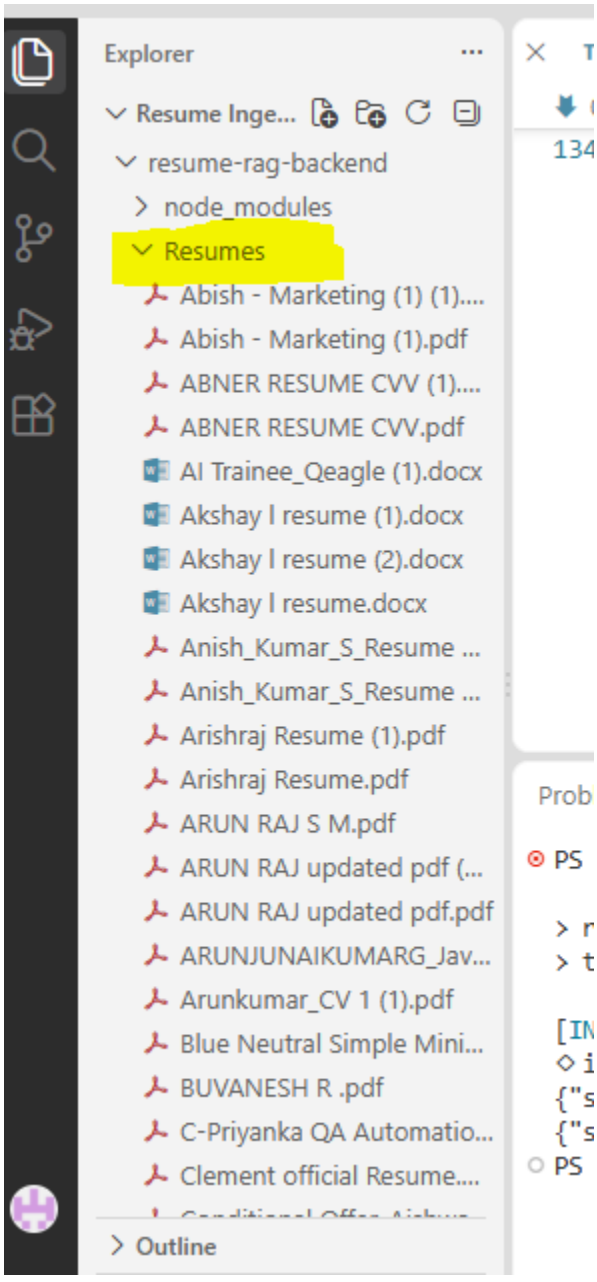
**Phase 2/Week5/Day2**

**Home Assignment : RAG_Ingestion_Resume**

1. Add the resume folder containing the resume files to the existing workspace.



2.  write a prompt to implement a batch ingestion workflow for resumes.

# Prompt: Resume Ingestion Workflow – Batch Processing Implementation Planning

Act as a Senior AI Architect and n8n Workflow Automation Engineer.

## Project Context

Review the existing application architecture and implementation completed up to **Phase 16**. Understand the current workflow, folder structure, ingestion pipeline, integrations, and existing functionality before proposing any changes.

A new folder named **Resumes** has been created, containing more than 160 resume files.

## Objective

Design an approach to execute the existing resume ingestion workflow efficiently using batch processing.

## Requirements

1. Analyze the existing architecture and identify the current resume ingestion workflow.
2. Verify how resume files are currently discovered, extracted, processed, and stored.

3. Identify the changes required to support batch execution without affecting existing functionality.
4. Configure the proposed batch-processing approach to handle:
    - Option 1: 5 resume files per batch.
    - Option 2: 10 resume files per batch.
5. Ensure that each batch is processed independently.
6. Avoid duplicate ingestion of previously processed resumes.
7. Maintain processing status and error logs for each resume.
8. Ensure failed files can be retried without reprocessing successful files.
9. Preserve existing integrations, database structures, APIs, and application functionality.
10. Provide batch execution progress, including total files, processed files, successful files, and failed files.

## Execution Planning

Before implementation, provide:

- Existing architecture analysis.
- Current ingestion workflow analysis.
- Proposed batch-processing architecture.
- Required workflow modifications.
- Expected number of batches for 160+ resumes.
- Error-handling and retry strategy.
- Duplicate detection strategy.
- Performance and resource considerations.
- Rollback and recovery approach.

## Strict Approval Requirements

**[STRICT RULE 1] DO NOT MODIFY ANY EXISTING CODE, FILES, WORKFLOWS, DATABASE RECORDS, OR CONFIGURATIONS.**

**[STRICT RULE 2] DO NOT START LIVE IMPLEMENTATION OR EXECUTE THE INGESTION PROCESS UNTIL I PROVIDE EXPLICIT APPROVAL.**

**[STRICT RULE 3] First provide the complete implementation plan and wait for my approval.**

**[STRICT RULE 4] The proposed workflow must support configurable batch sizes of 5 or 10 resume files per execution.**

**[STRICT RULE 5] Do not trigger all 160+ resumes for processing in a single execution.**

## Expected Deliverables

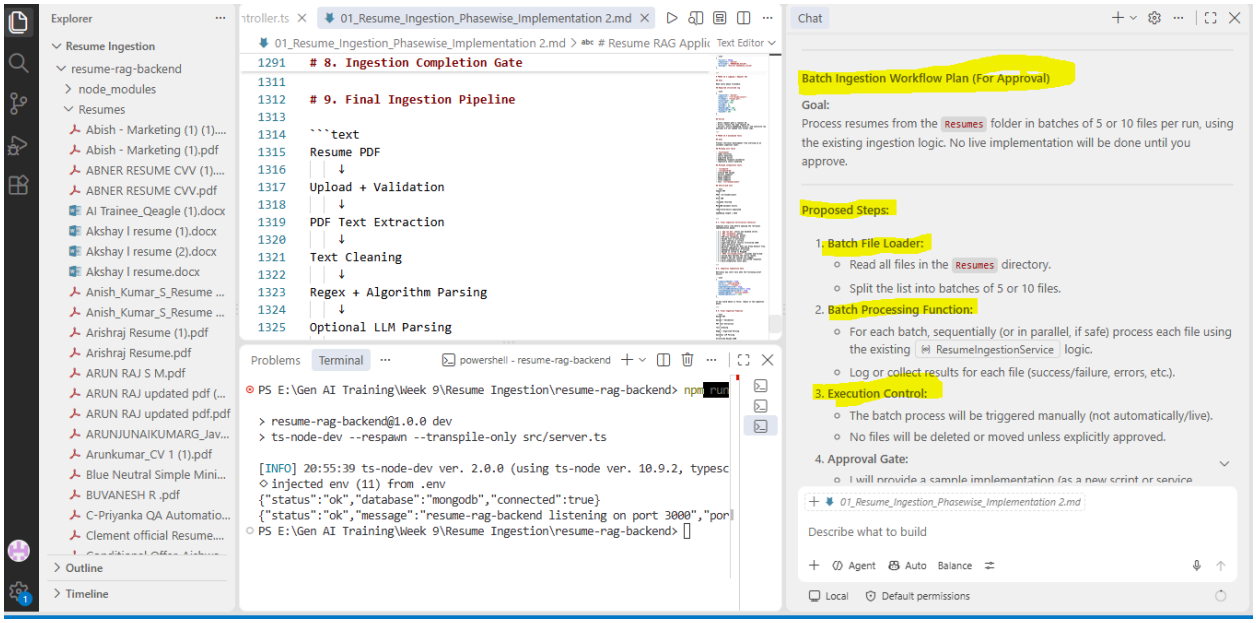
1. Existing architecture assessment.
2. Proposed batch ingestion workflow diagram.
3. Phase-wise implementation plan.
4. Batch execution strategy (5 or 10 files).
5. Error handling and retry mechanism.
6. Duplicate prevention mechanism.
7. Execution monitoring and reporting approach.
8. Risk assessment.

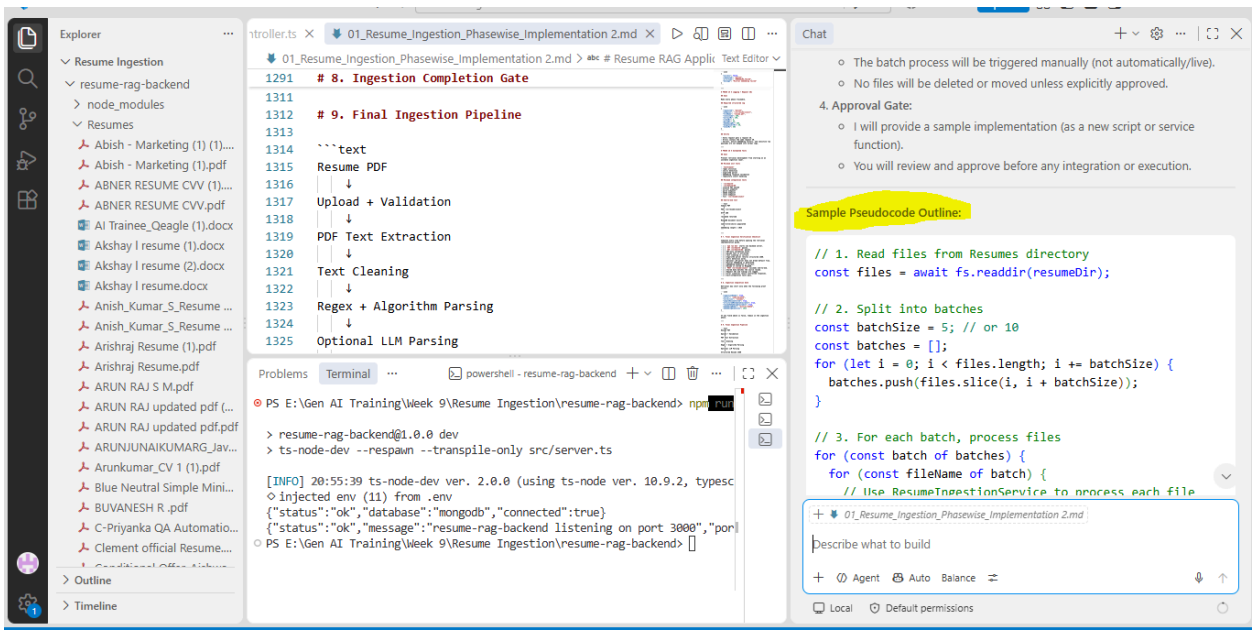
## Final Instruction

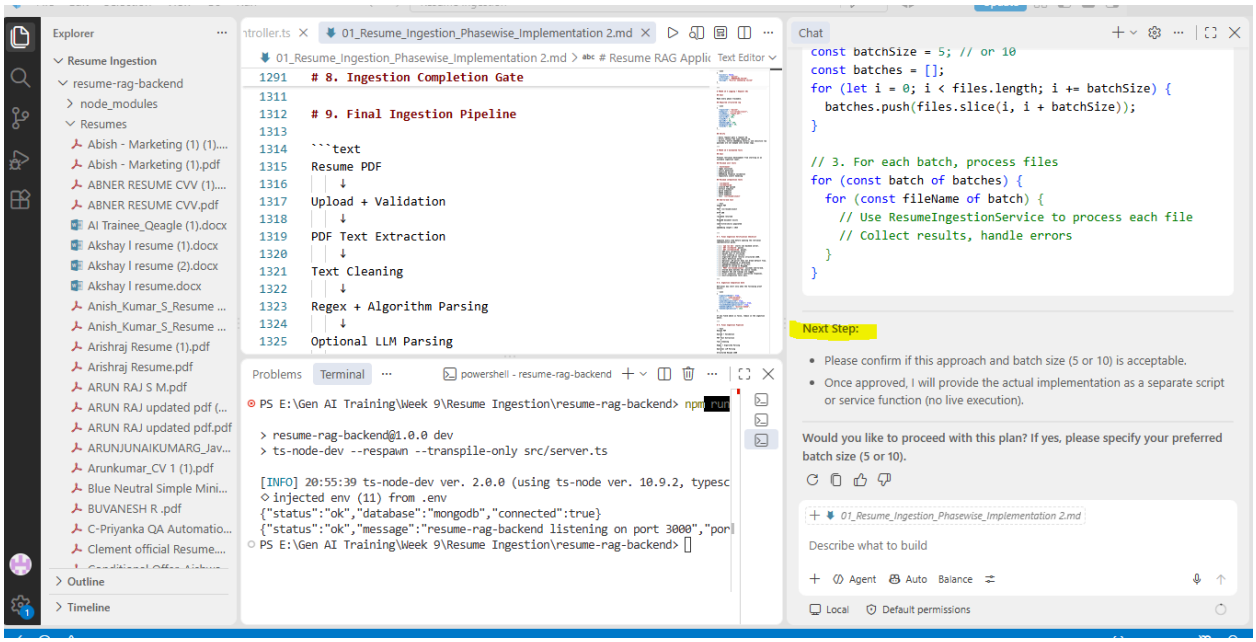
First analyze the existing implementation up to Phase 16 and present your findings and recommendations.

**STOP AFTER PROVIDING THE IMPLEMENTATION PLAN. WAIT FOR MY EXPLICIT APPROVAL BEFORE MAKING ANY CHANGES OR EXECUTING ANY WORKFLOW.**

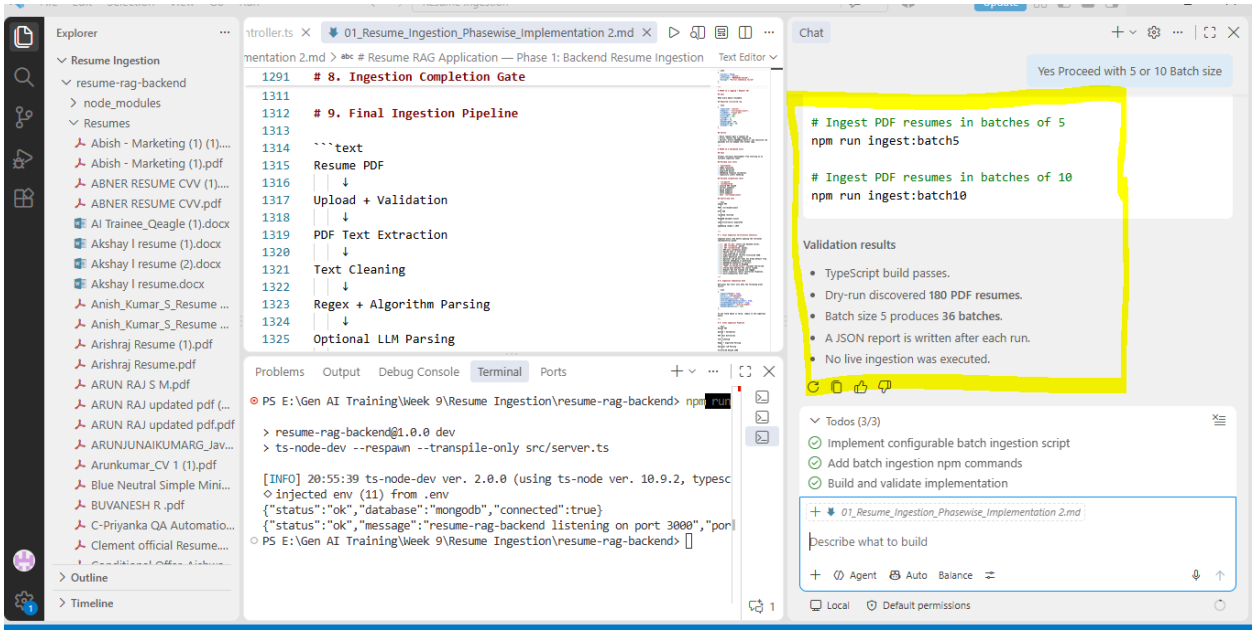
3. Configure the workflow to process resumes in batches of 5 or 10 files per run.

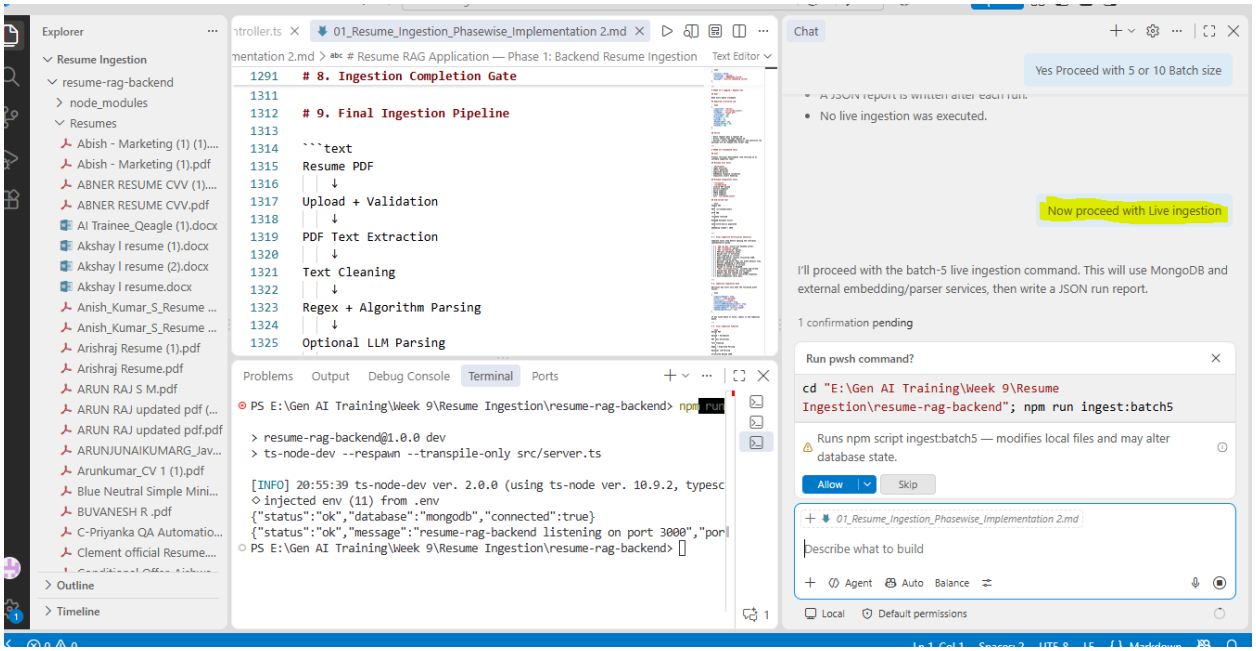


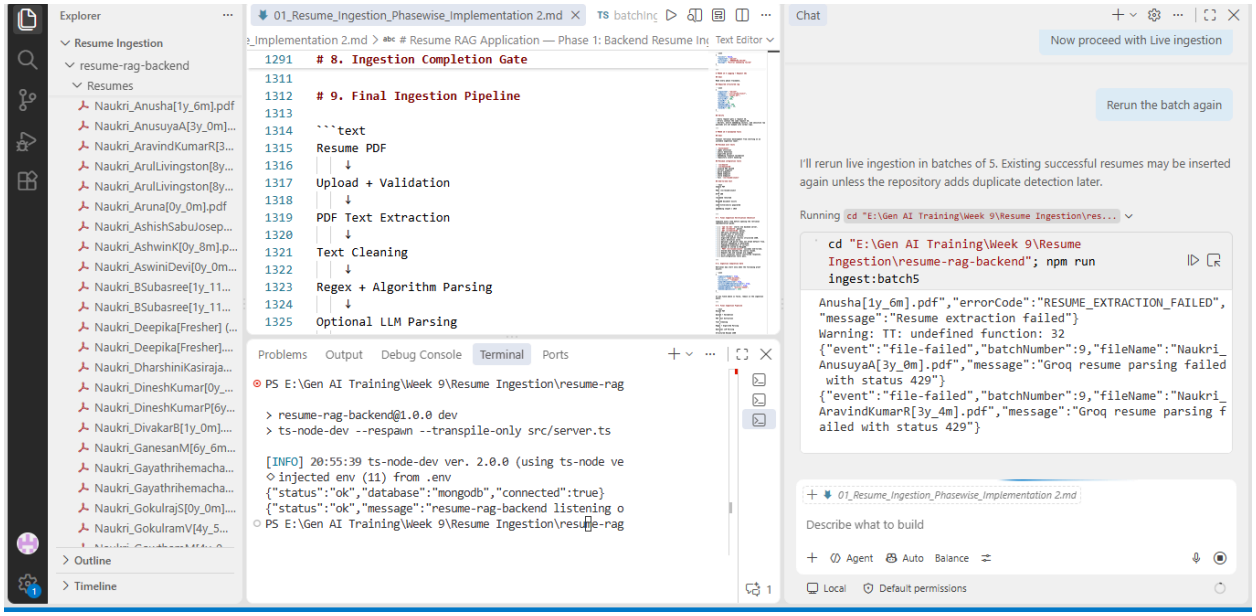



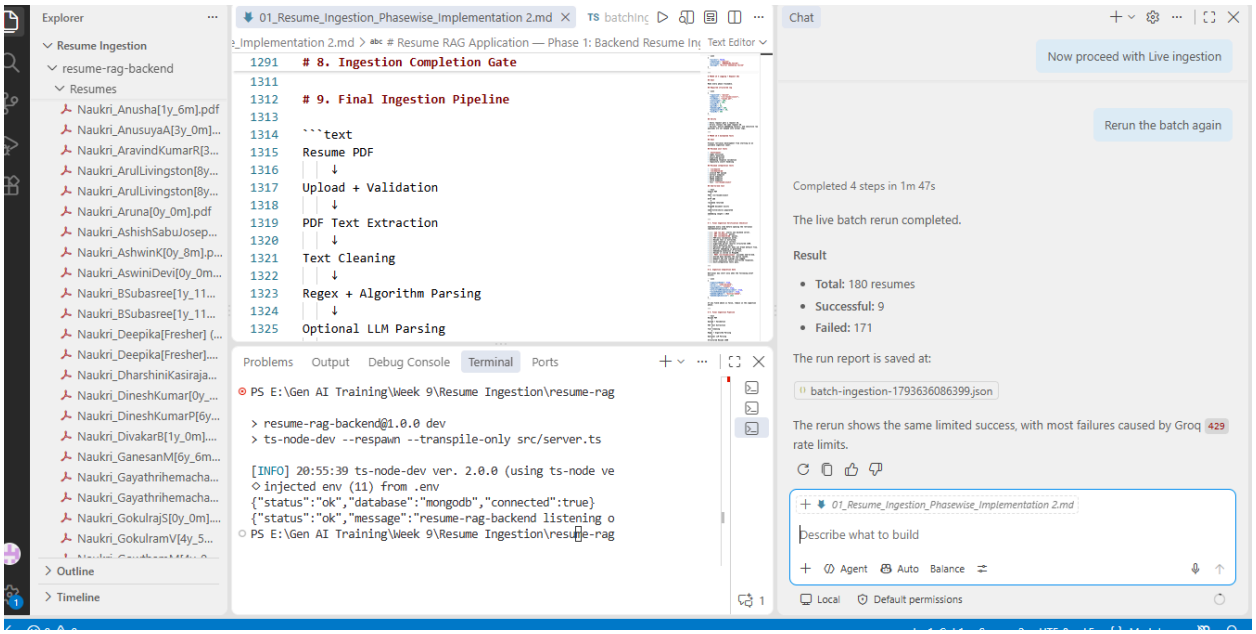



4. Validate the ingestion process by testing multiple batches and verifying that all resumes are successfully ingested.

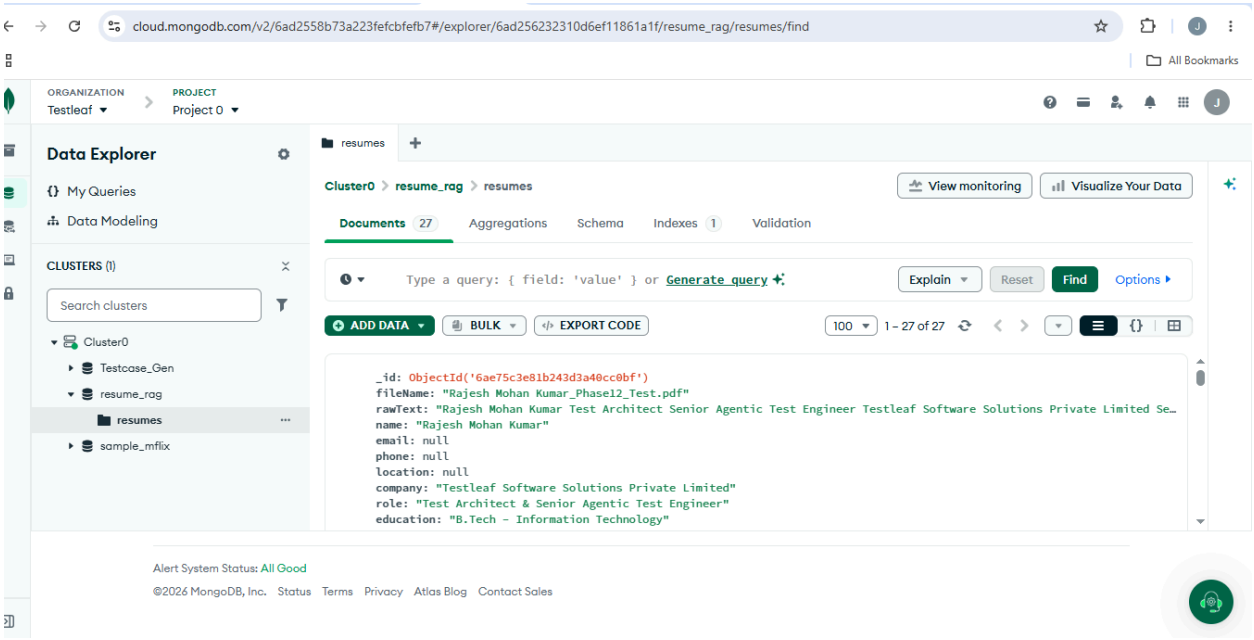

Explorer
Resume Ingestion
resume-rag-backend
node_modules
Resumes
01_Resume_Ingestion_Phasewise_Implementation 2.md
# 8. Ingestion Completion Gate
# 9. Final Ingestion Pipeline
```text
Resume PDF
↓
Upload + Validation
↓
PDF Text Extraction
↓
Text Cleaning
↓
Regex + Algorithm Parsing
↓
Optional LLM Parsing
Problems
Output
Debug Console
Terminal
Ports
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typesc
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","por
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Chat
Yes Proceed with 5 or 10 Batch size
# Ingest PDF resumes in batches of 5
npm run ingest:batch5
# Ingest PDF resumes in batches of 10
npm run ingest:batch10
Validation results
TypeScript build passes.
Dry-run discovered 180 PDF resumes.
Batch size 5 produces 36 batches.
A JSON report is written after each run.
No live ingestion was executed.
Todos (3/3)
Implement configurable batch ingestion script
Add batch ingestion npm commands
Build and validate implementation
Describe what to build
Agent
Auto
Balance
Local
Default permissions


Explorer
Resume Ingestion
01_Resume_Ingestion_Phasewise_Implementation 2.md
Problems
Output
Debug Console
Terminal
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PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run
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injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","por
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Chat
Yes Proceed with 5 or 10 Batch size
No live ingestion was executed.
Now proceed with Live ingestion
I'll proceed with the batch-5 live ingestion command. This will use MongoDB and external embedding/parser services, then write a JSON run report.
1 confirmation pending
Run pwsh command?
cd "E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend"; npm run ingest:batch5
Runs npm script ingest:batch5 — modifies local files and may alter database state.
Allow
Skip
Describe what to build
Agent
Auto
Balance
Local
Default permissions
```




Once the batch process started , parallel check in Mongo DB for Ingestion


cloud.mongodb.com/v2/6ad2558b73a223fefcbfefb7#/explorer/6ad256232310d6ef11861a1f/resume_rag/resumes/find
All Bookmarks
ORGANIZATION
Testleaf
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
Testcase_Gen
resume_rag
resumes
sample_mflix
resumes
Cluster0 > resume_rag > resumes
View monitoring
Visualize Your Data
Documents 27
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
100
1 – 27 of 27
_id: ObjectId('6ae75c3e81b243d3a40cc0bf')
fileName: "Rajesh Mohan Kumar_Phase12_Test.pdf"
rawText: "Rajesh Mohan Kumar Test Architect Senior Agentic Test Engineer Testleaf Software Solutions Private Limited Se…
name: "Rajesh Mohan Kumar"
email: null
phone: null
location: null
company: "Testleaf Software Solutions Private Limited"
role: "Test Architect & Senior Agentic Test Engineer"
education: "B.Tech – Information Technology"
Alert System Status: All Good
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